RISK 201: Asset-Liability Management

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Assets and liabilities

- Insurance company **liabilities** consist of future cash flows promised to policyholders (e.g., annuity or pension payments, payouts on settlements after an injury, etc.).
- These liabilities must be backed by **assets**.
- The best assets to back a liability stream are ones that produce cash flows with the same timing and in the same amount as the liabilities. This is called exact cash flow matching or dedication.

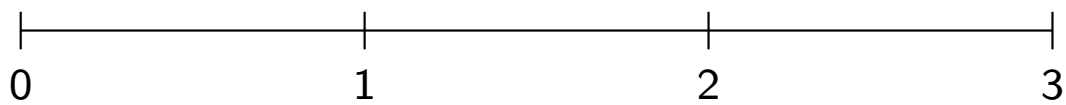
Exact cash flow matching

- Creating a dedicated portfolio using zero-coupon bonds is easy $\rightarrow$ buy bonds to match CFs one by one
- When using bonds with coupons:
  1. Consider the final liability cash flow first.
  2. Find the amount of the asset needed to match the last liability cash flow.
  3. Determine the liability cash flows that are left over after the first matching asset is chosen.
  4. Consider the longest remaining cash flow; repeat steps 2–4 until all liability cash flows are matched.

Example 9.1

A company has an obligation to pay \$10,000 at the end of each year for 3 years. Spot rates are 4%, 5%, and 5.8% for terms of 1, 2 and 3 years, respectively.

a) Construct a dedicated portfolio to match the cash flows using 1-, 2- and 3-year zero coupon bonds.



1-year zero:

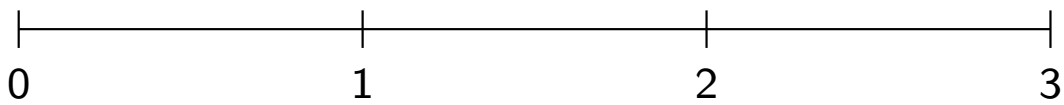
2-year zero:

3-year zero:

Example 9.1, continued

b) Construct a dedicated portfolio using the following:

- Bond 1: 3-year bond with 5% annual coupons
- Bond 2: 2-year bond with 2% annual coupons
- Bond 3: 1-year zero-coupon bond



3-year bond:

2-year bond:

1-year zero:

Interest rate sensitivity

- Creating dedicated portfolios is not always possible; assets and liabilities might not be perfectly matched.
- Even if we pick assets whose PV matches the PV of the liabilities, the PVs might drift apart when interest rates change and
  - the value of fixed-income assets, like bonds, changes
  - the (market-consistent) value of the liabilities changes
- Goal: choose assets so that the PV of the assets moves in unison with the PV of the liabilities
  $\rightarrow$ protect surplus from changes in interest rates

Surplus

- Define the surplus, $S(i)$, as the PV of assets minus the PV of liabilities, all valued at rate i:

$$S(i) = \sum_{t=0}^{n} \frac{A_t}{(1+i)^t} - \sum_{t=0}^{n} \frac{L_t}{(1+i)^t}$$

- Select a portfolio so that $S(i) > 0$ under certain conditions. This is called immunization.

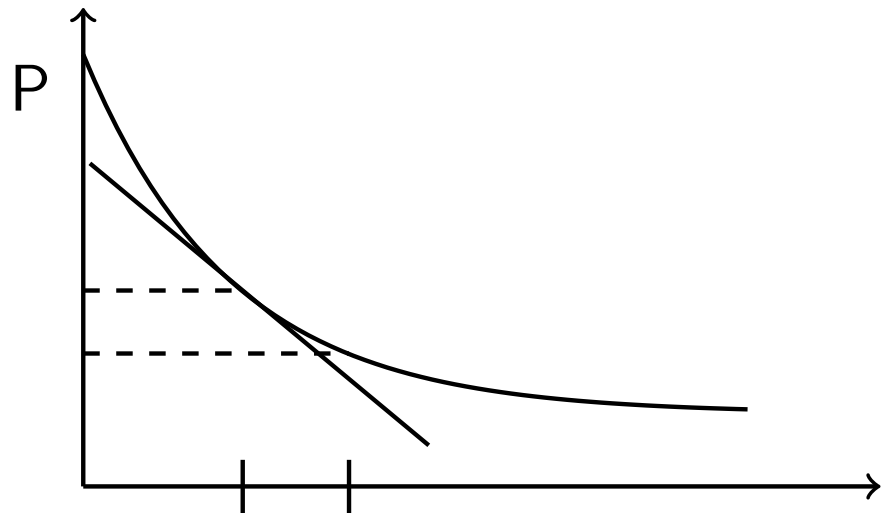
- To do so, we need to understand how the PV of cash flows change in response to changes in interest rates.

9.1 Duration

- Given annual cash flows C_t and continuously compounded yield δ, the price at time 0 is

$$P(\delta) = \sum_{t=0}^{n} C_t \, e^{-\delta t}$$

- Price curve is usually decreasing and convex.



Approximating the price change

- The price change resulting from a change in the continuously compounded yield is

$$P(\delta + \Delta\delta) = P(\delta) + (\Delta\delta)P'(\delta) + \frac{(\Delta\delta)^2 P''(\delta)}{2!} + \cdots$$

- A first-order approximation gives

$$P(\delta + \Delta\delta) \approx P(\delta) + (\Delta\delta)P'(\delta)$$

- Graphically, this is a linear approximation, based on the current price, $P(\delta)$, and the tangent to the price curve at δ.

Percentage change in price

- For bonds, we are often interested in the percentage change in the price in response to a change in the (continuously compounded) yield δ:

$$\%\Delta P(\delta) = \frac{\Delta P(\delta)}{P(\delta)} = \frac{P(\delta + \Delta\delta) - P(\delta)}{P(\delta)} = (\Delta\delta)\,\frac{P'(\delta)}{P(\delta)}$$

Duration

(Macaulay) duration is the negative of the derivative of the price function with respect to the continuously compounded yield, divided by the price:

$$\bar{d} = -\frac{\frac{d}{d\delta}P(\delta)}{P(\delta)}$$

If we know the cash flows' duration, we can easily approximate the % change in price from a change in δ:

$$\%\Delta P(\delta) \approx (\Delta\delta)\,\frac{P'(\delta)}{P(\delta)} = -\bar{d}\,(\Delta\delta)$$

Measuring the change in yield

- The change in the yield is measured in absolute terms. We often express this in basis points:
  1 basis point (bp) $= 0.01\%$ $\quad$ (100 bps $= 1\%$)
- When we say a rate is x bps higher/lower than some other rate, we always mean this additively.
- Example 9.2: Suppose the 30-year continuously compounded yield observed in the market on Apr 1 is 10%. Determine the yield on April 15 if you are told it is 5 bps higher than the rate on April 1.

Interpreting (Macaulay) duration

$$P(\delta) = \sum_{t=0}^{n} C_t\, e^{-\delta t} \;\Rightarrow\; \frac{d}{d\delta} P(\delta) = \sum_{t=0}^{n} -t \cdot C_t\, e^{-\delta t}$$

$$\bar{d} = \frac{\sum_{t=0}^{n} t \cdot C_t\, e^{-\delta t}}{\sum_{t=0}^{n} C_t\, e^{-\delta t}} = \sum_{t=0}^{n} t \cdot \left(\frac{C_t\, e^{-\delta t}}{\sum_{t=0}^{n} C_t\, e^{-\delta t}} \right) = \sum_{t=0}^{n} t \cdot \frac{PV_t}{PV_{all}}$$

Macaulay duration is the weighted average of the times that cash flows occur, where the weights are the relative contributions of each cash flow to the price.

Observations

1. The (Macaulay) duration of an n-year zero-coupon bond is n.
2. A bond with a higher coupon rate has a shorter Macaulay duration because the higher coupons mean more weight is placed on earlier payments and less weight on late payments.
3. A bond with a higher yield has a shorter Macaulay duration because the relative contribution of the later cash flows to the price is smaller when the yield is higher, so these cash flows have a smaller weight.

Example 9.3

A 4-year bond has 6% annual coupons and is priced at par.

a) Find the duration of this bond.

t	C_t	$C_t e^{-\delta t}$	$t \cdot C_t e^{-\delta t}$
1	$0.06F$	$0.06F/(1.06)$	$1 \cdot 0.06F/(1.06)$
2	$0.06F$	$0.06F/(1.06)^2$	$2 \cdot 0.06F/(1.06)^2$
3	$0.06F$	$0.06F/(1.06)^3$	$3 \cdot 0.06F/(1.06)^3$
4	$1.06F$	$1.06F/(1.06)^4$	$4 \cdot 1.06F/(1.06)^4$

Example 9.3, continued

b) Approximate the percentage change in price resulting from a 10 bps increase in the continuously compounded yield.

9.2 Modified duration (volatility)

- Yield is usually expressed as an effective rate of interest per period, i, not a cont. comp. rate δ.
- We are normally interested in knowing/predicting the percentage change in price on account of a change in the yield i (not δ).
- Modified duration (aka volatility) is defined as the negative derivative of the price function with respect to the effective yield, divided by the price:

$$\bar{v} = -\frac{\frac{d}{di}P(i)}{P(i)}$$

Connection between $\bar{d}$ and $\bar{v}$

- With annual effective rate i:

$$P(i) = \sum_{t=0}^{n} C_t (1+i)^{-t} \Rightarrow \frac{d}{di} P(i) = \sum_{t=0}^{n} -t \cdot C_t (1+i)^{-t-1}$$

$$\bar{v} = \frac{\sum_{t=0}^{n} t \cdot C_t (1+i)^{-t-1}}{\sum_{t=0}^{n} C_t (1+i)^{-t}} = \frac{\bar{d}}{1+i}$$

- When i is a semi-annual rate: $\bar{v} = \dfrac{\bar{d}/2}{1+i}$
- When i is an m-thly rate: $\bar{v} = \dfrac{\bar{d}/m}{1+i}$

Approximating the change in price

If we know the cash flows' volatility, we can easily approximate the change in price due to a change in the effective yield, i:

$$\%\Delta P(i) \approx (\Delta i)\,\frac{\frac{d}{di}P(i)}{P(i)} = -\bar{v}\,(\Delta i)$$

$$P(i+\Delta i) \approx P(i)\cdot[1-\bar{v}\,(\Delta i)]$$

Example 9.4

Consider the same bond as in Example 9.3: a 4-year bond with 6% annual coupons priced at par.

a) Find the modified duration (volatility) of this bond.

b) What is the approximate percentage change in the price of this bond in response to a 75 bps increase in the effective yield?

Example 9.4, continued

c) What is the exact percentage change in the price of this bond in response to a 75 bps increase in the effective yield?

How good was the approximation in part (b)?

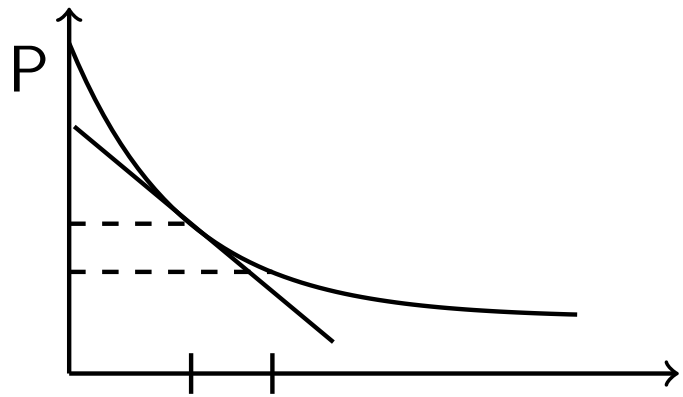
Portfolio duration

- Suppose a portfolio consists of n assets, with n separate series of annual cash flows. Using an effective annual interest rate of i, asset k has price $P_k(i)$ and duration D_k.
- The total price of the portfolio is $P(i) = \sum_{k=1}^{n} P_k(i)$ and the duration of the portfolio is

$$D = -\frac{P'(i)}{P(i)} = -\frac{\sum_{k=1}^{n} P'_k(i)}{P(i)} = -\sum_{k=1}^{n} \frac{P'_k(i)}{P_k(i)} \frac{P_k(i)}{P(i)} = \sum_{k=1}^{n} D_k \frac{P_k(i)}{P(i)}$$

9.3 Convexity

- Modified duration works well to estimate the price change for small changes in i ($\sim$less than 50 bps).
- For larger changes in yield, we need to take into account the convexity of the price function.



Second-order approximation

- The price change resulting from a change in the annual effective yield is

$$P(i+\Delta i) = P(i) + (\Delta i)P'(i) + \frac{(\Delta i)^2 P''(i)}{2!} + \cdots$$

- A second-order approximation gives

$$P(i+\Delta i) \approx P(i)\left[1 - (\Delta i)\bar{v} + \frac{1}{2}(\Delta i)^2 \frac{P''(i)}{P(i)}\right]$$

Convexity

Define the convexity, $\bar{c}$, of a set of cash flows as the second derivative of the price function with respect to the effective yield, divided by the price

$$P(i) = \sum_{t=0}^{n} C_t (1+i)^{-t} \Rightarrow \frac{d}{di} P(i) = \sum_{t=0}^{n} -t \cdot C_t (1+i)^{-t-1}$$

$$\frac{d^2}{di^2} P(i) = \sum_{t=0}^{n} t(t+1) \cdot C_t (1+i)^{-t-2}$$

$$\bar{c} = \frac{P''(i)}{P(i)} = \frac{\sum_{t=0}^{n} t(t+1) \cdot C_t (1+i)^{-t}}{(1+i)^2 P(i)}$$

Example 9.5

Consider the same bond as in Example 9.4: a 4-year bond with 6% annual coupons, priced at par.

a) Find the convexity of this bond.

t	C_t	$t(t+1) \cdot C_t(1+i)^{-(t+2)}$
1	$0.06F$	$1(2) \cdot 0.06F/(1.06)^3$
2	$0.06F$	$2(3) \cdot 0.06F/(1.06)^4$
3	$0.06F$	$3(4) \cdot 0.06F/(1.06)^5$
4	$1.06F$	$4(5) \cdot 1.06F/(1.06)^6$

Example 9.5, continued

b) What is the approximate percentage change in the price of this bond in response to a 75 bps increase in the effective yield, using convexity?

How good is this approximation?

Portfolio convexity

The convexity of a portfolio is the weighted average of the convexity of each item in the portfolio, with the weight being the price of the item in proportion to the price of the entire portfolio:

$$C = \sum_{k=1}^{n} \bar{c}_k \frac{P_k(i)}{P(i)}$$

9.4 Immunization

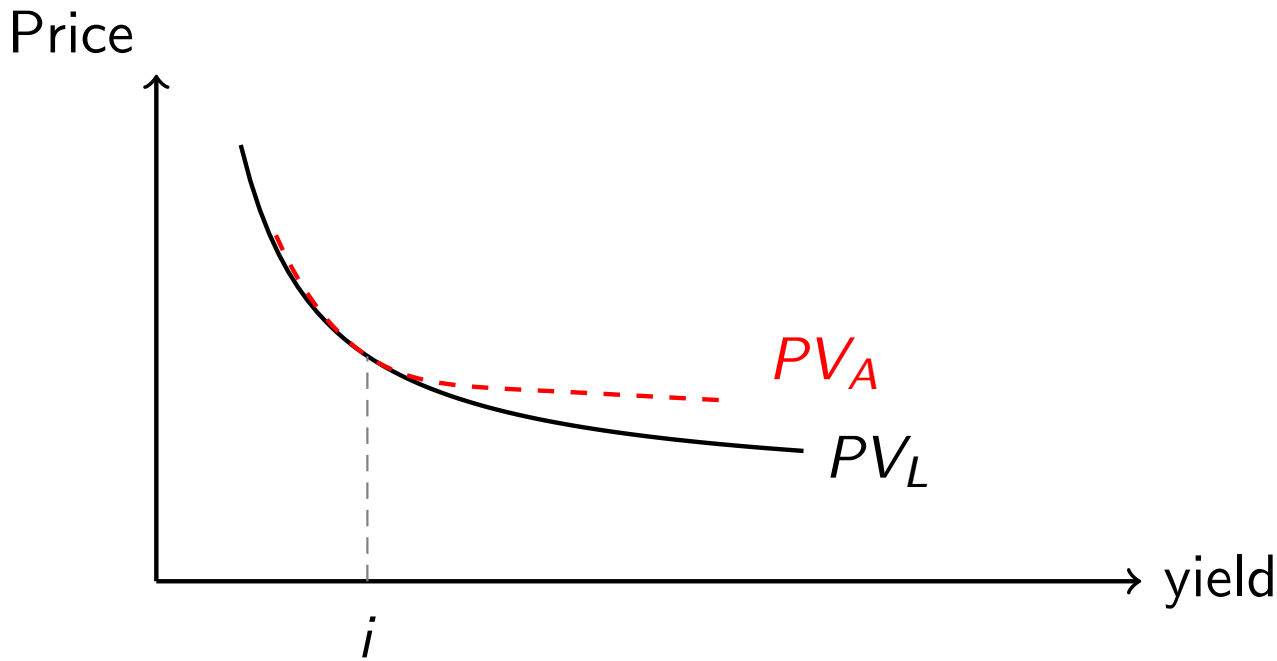
- Suppose we have a series of liability cashflows and we are trying to construct an asset portfolio so that the surplus (PV of assets − PV of liabilities) does not become negative when interest rates change
- Two choices:
  1. Redington immunization
  2. Full immunization

Redington immunization

- Three conditions:
  1. $PV_A(i) \geq PV_L(i)$
  2. $\bar{d}_A(i) = \bar{d}_L(i)$
  3. $\bar{c}_A(i) > \bar{c}_L(i)$

Redington immunization, cont'd

- Ensures that $PV_A(i) \geq PV_L(i)$ for any small shift in the yield rate



Assumptions (Redington imm'n)

- Flat yield curve
- Parallel shifts in the yield curve (yields change by the same amount for all maturities)

Example 9.6

An insurance company has committed to making a payment of \$100,000 in 10 years.

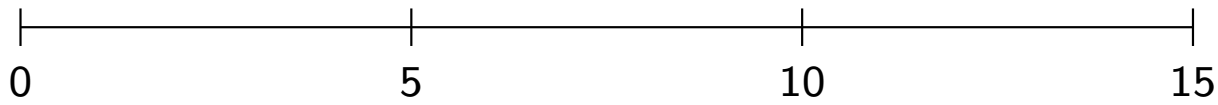
In order to fund its liability, the company has invested \$27,919.74 in a 5-year zero-coupon bond and \$27,919.74 in a 15-year zero-coupon bond.

The annual effective yield is 6%.

Determine whether the company's position is immunized.

Example 9.6, continued

Liability CF:



Asset CF:

Full immunization

- A portfolio is fully immunized at i if

$$S(i+\varepsilon) \geq 0 \text{ for certain } \varepsilon > 0$$

- Assuming a flat yield curve and parallel shifts in yields, a single cash flow is fully immunized if
  1. $PV_A(i) = PV_L(i)$
  2. $\bar{d}_A(i) = \bar{d}_L(i)$
  3. Asset cash flows occur both before and after the liability cash flow

Full immunization, continued

Cash flows:



where $PV_{A_1}(i) = PV_{A_2}(i)$.

	Liability	**Asset portfolio**
Duration	t	t
Convexity	$t(t+1)$	$> t(t+1)$ (why?)

Full immunization, continued

- To fully immunize a series of liability cash flows, choose assets to fully immunize each liability cash flow separately.
- Some practical issues:
  - Yield curves are usually not flat
  - Changes in yield curve are not parallel
  - Asset cash flows contain risk
  - Uncertainty in liability cash flows

Example 9.7

Liability payments of 100 each are due to be paid in 2, 4 and 6 years from now. Asset cash flow consists of A_1 in 1 year and A_5 in 5 years. The yield for all payments is 10%.

An attempt is made to have the asset cash flow immunize the liability cash flow by matching present value and duration.

Example 9.7, continued

a) Find A_1 and A_5.

Example 9.7, continued

b) Determine whether the conditions for Redington immunization are satisfied.